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Analog IC Design Track
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Lab 09 (Mini Project 01)
Two-Stage Miller OTA

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1 PART 1: OTA Design

1.1 Target Specifications

Parameter	Target Specification (0.18 μm)
Technology	0.18 μm
Supply Voltage (V_{DD})	1.8 V
Static Gain Error	$\leq 0.05\%$
DC Open-Loop Gain	≥ 66 dB (Derived)
Phase Margin (PM)	$\geq 70^\circ$
CMRR @ DC	≥ 74 dB
OTA Current Consumption	≤ 60 μA
CMIR - low	≤ 0.2 V
CMIR - high	≥ 1.0 V
Output Swing	0.2 V to 1.6 V
Load Capacitance (C_L)	5 pF
Buffer Rise Time (10% to 90%)	≤ 70 ns
Slew Rate (SR)	5 V/ μs

1.2 Topology Selection

For this design, a Two-Stage Operational Transconductance Amplifier (OTA) topology was selected to simultaneously achieve high DC gain and a wide output voltage swing. A PMOS input differential pair was specifically chosen over an NMOS pair to satisfy the Common-Mode Input Range (CMIR) requirement. Since the required CMIR extends close to ground (0 V), the PMOS input transistors can remain in the saturation region even at very low input common-mode voltages, which would otherwise drive an NMOS input pair into the cut-off or triode regions. The second stage consists of a common-source amplifier to provide the necessary additional voltage gain and maximize the output swing.

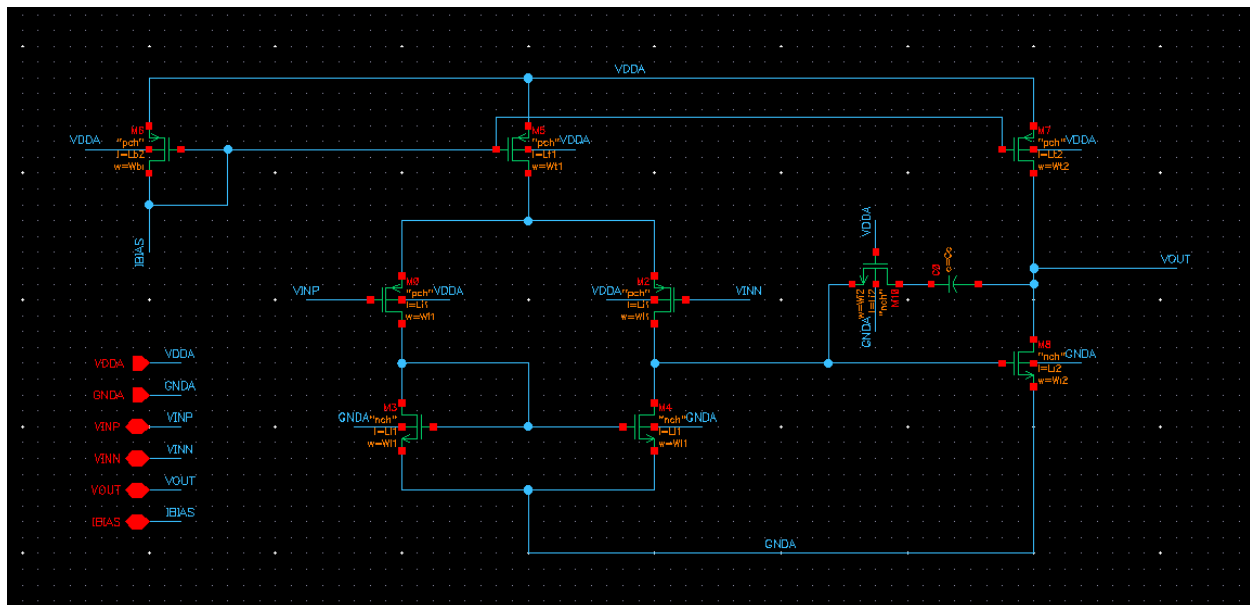


Figure 1: two stage _OTA

1.3 Design Methodology and Circuit Sizing

1.3.1 Input Differential Pair Sizing (M_0 , M_2)

The sizing of the PMOS input differential pair was executed systematically based on the project's dynamic specifications and the suggested design procedure. First, the compensation capacitor (C_C) was initialized to half the load capacitance ($0.5C_L$), yielding $C_C = 2.5$ pF.



The required unity gain frequency (ω_u) was determined from the closed-loop buffer rise time specification ($t_{rise} \leq 70$ ns). Utilizing the first-order approximation $t_{rise} = 2.2/\omega_u$, the minimum required bandwidth is: $\omega_u \geq \frac{2.2}{70 \text{ ns}} \Rightarrow \omega_u \geq 31.428 \text{ Mrad/s}$

From this, the transconductance for the input pair ($g_{m1,2}$) was calculated:

$$g_{m1,2} = \omega_u \times C_C \geq 78.5 \text{ uS}$$

Simultaneously, the total first-stage bias current (I_{B1}) was dictated by the Slew Rate (SR) requirement of 5 V/ μs . Using the relation $SR = I_{B1}/C_C$, the tail current was calculated as $I_{B1} = 12.5 \text{ }\mu\text{A}$. This sets the branch current for each input transistor to $I_{D1,2} = 6.25 \text{ }\mu\text{A}$.

With both $g_{m1,2}$ and $I_{D1,2}$ established, the minimum required transconductance efficiency was computed:

$$\frac{g_{m1,2}}{I_{D1,2}} \geq \frac{78.5 \text{ uS}}{6.25 \text{ }\mu\text{A}} = 12.56 \text{ V}^{-1}$$

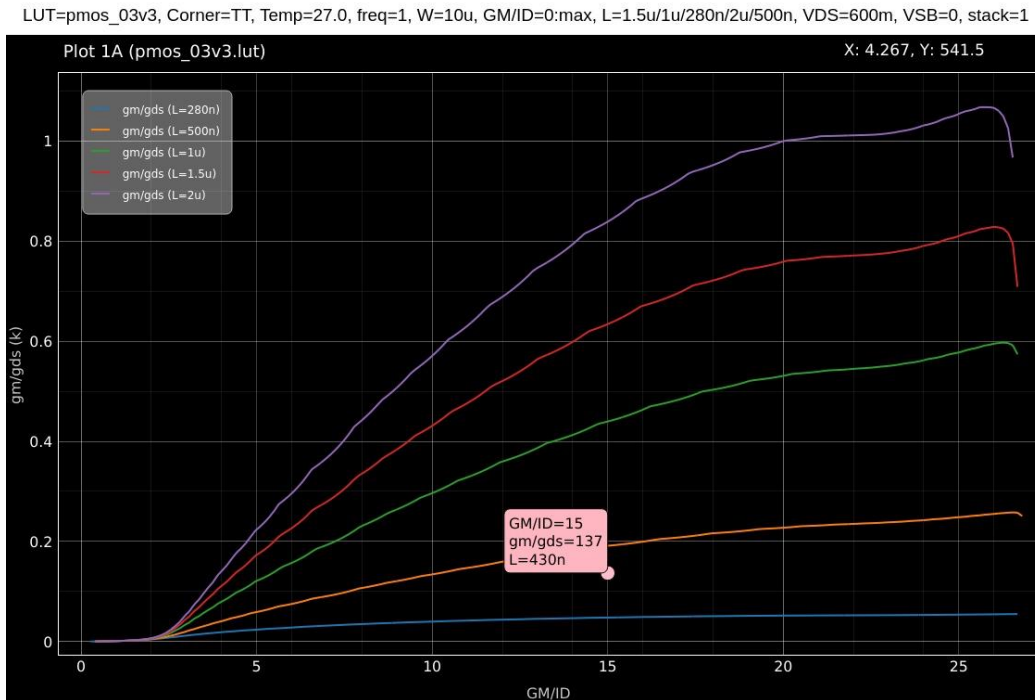


Figure 2: gm/id vs. gm/gds input

LUT=pmos_03v3, Corner=TT, Temp=27.0, freq=1, W=10u, GM/ID=0:max, L=1.5u/1u/280n/2u/500n, VDS=600m, VSB=0, stack=1

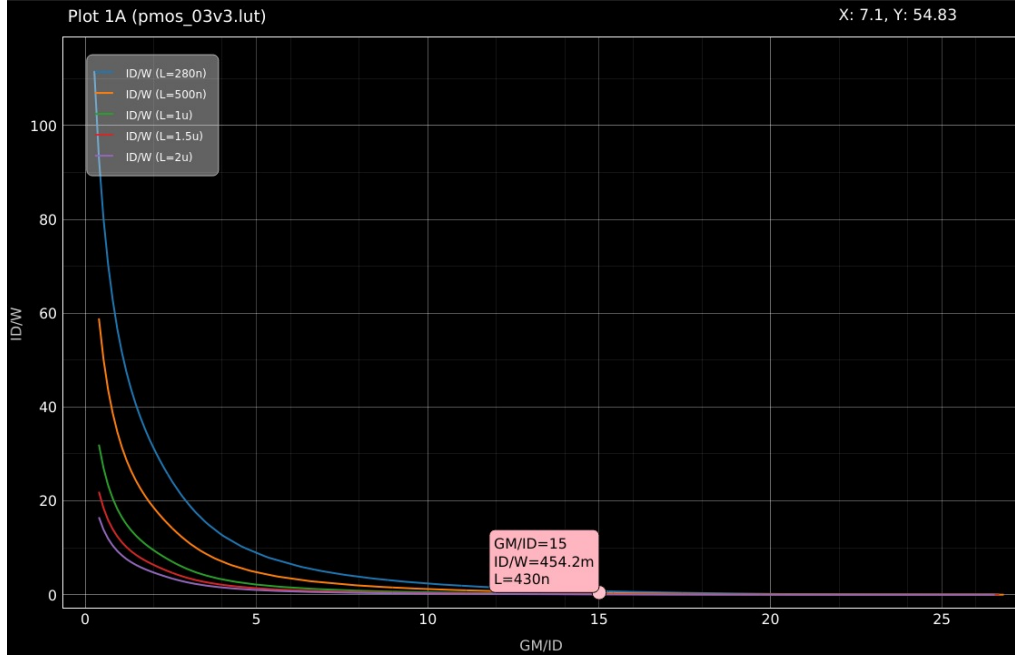


Figure 3: ID/W input pair

To provide a sufficient design margin and ensure a high intrinsic gain for the first stage, a g_m/I_D ratio of 15 V^{-1} was selected. Based on the ADT lookup tables for this specific operating point, a channel length of $L_{1,2} = 430 \text{ nm}$ was chosen. At this length and g_m/I_D , the corresponding current density is $I_D/W = 454.2 \text{ mA/m}$. Finally, the width of the input transistors was calculated as: $W_{1,2} = \frac{6.25 \text{ uA}}{454.2 \text{ mA/m}} = 13.76 \text{ um}$

1.3.2 First-Stage Current Mirror Load Sizing (M_3, M_4)

The NMOS current mirror load (M_3, M_4) of the first stage was sized following the suggested design procedure, which recommends assuming a relatively large g_m/I_D (e.g., 15 V^{-1}) to obtain an initial estimate for the channel length. This ensures a high Early voltage (V_A) and consequently a sufficient intrinsic gain contribution from the load.



LUT=nmos_03v3, Corner=TT, Temp=27.0, freq=1, W=40u, GM/ID=0:max, L=1.5u/1u/280n/2u/500n, VDS=600m, VSB=0, stack=1

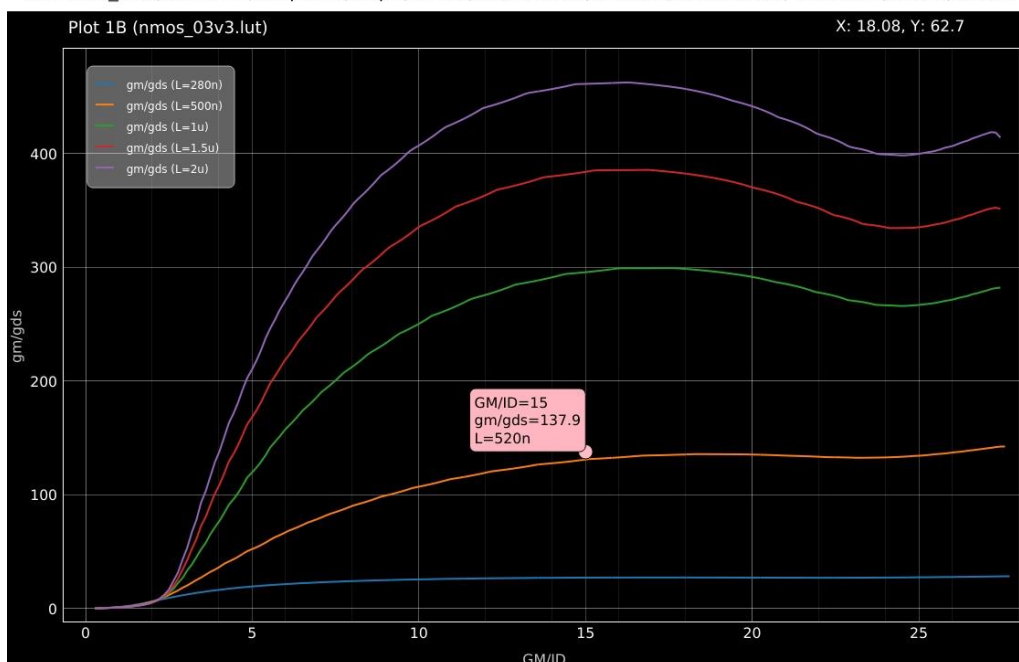


Figure 4: gm/gds for current mirror

LUT=nmos_03v3, Corner=TT, Temp=27.0, freq=1, W=40u, GM/ID=0:max, L=1.5u/1u/280n/2u/500n, VDS=600m, VSB=0, stack=1

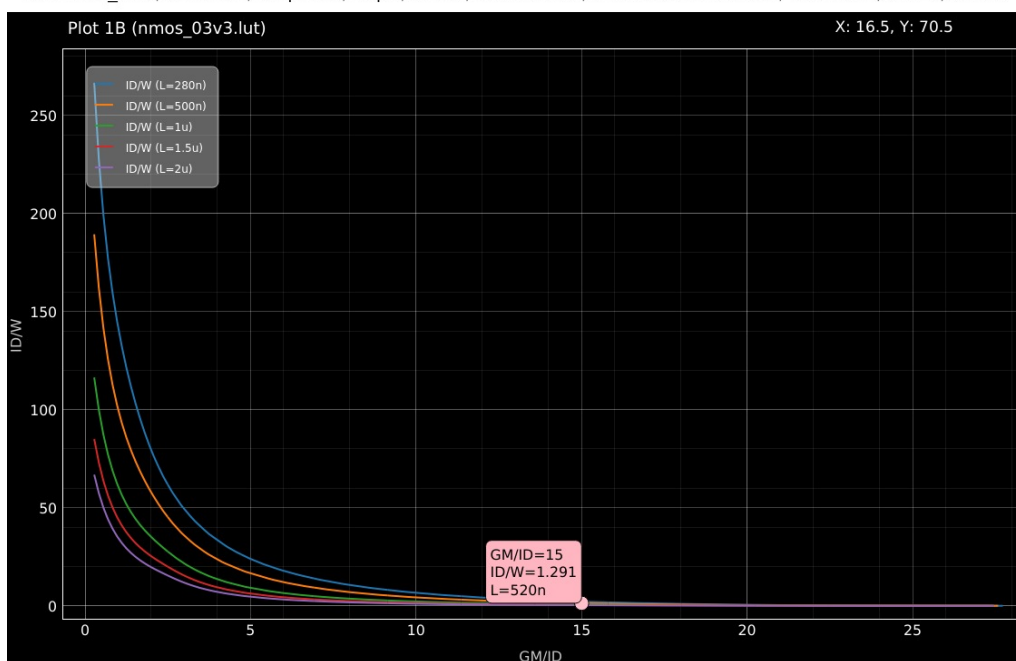


Figure 5: ID/W for current mirror

Assuming $g_m/I_D = 15 \text{ V}^{-1}$, the NMOS lookup tables (LUTs) generated by ADT were utilized to determine the appropriate dimensions. To maintain a high first-stage DC gain, the g_m/g_{ds} versus g_m/I_D design chart was inspected. A channel length of $L_{3,4} = 520 \text{ nm}$ was selected, which yields an intrinsic gain of $g_m/g_{ds} = 137.9 \text{ V/V}$ at the specified operating point.

The drain current flowing through each load transistor is identical to the input pair branch current, meaning $I_{D3,4} = 6.25 \text{ }\mu\text{A}$. With the length and g_m/I_D established, the current density was extracted from the I_D/W versus g_m/I_D chart. At $L = 520 \text{ nm}$ and $g_m/I_D = 15 \text{ V}^{-1}$, the current density is $I_D/W = 1.291 \text{ A/m}$. Finally, the required width for the current mirror load transistors was calculated by dividing the branch current by the extracted current density:

$$W_{3,4} = \frac{6.25 \text{ }\mu\text{A}}{1.291 \text{ A/m}} = 4.84 \text{ }\mu\text{m}$$

1.3.3 Tail Current Source (M_5) and Biasing Network

The sizing of the PMOS tail current source (M_5) was driven by the required Common-Mode Input Range (CMIR-high) and the output voltage swing specifications. To ensure that M_5 remains in the saturation region even when the input common-mode voltage reaches its maximum limit, a restrictive overdrive voltage limit was applied. The maximum $V_{DS,sat}$ for the tail current source was estimated as 0.2 V . Setting the characteristic voltage V^* equal to this saturation voltage ($V^* = 0.2 \text{ V}$) introduces a necessary design margin.

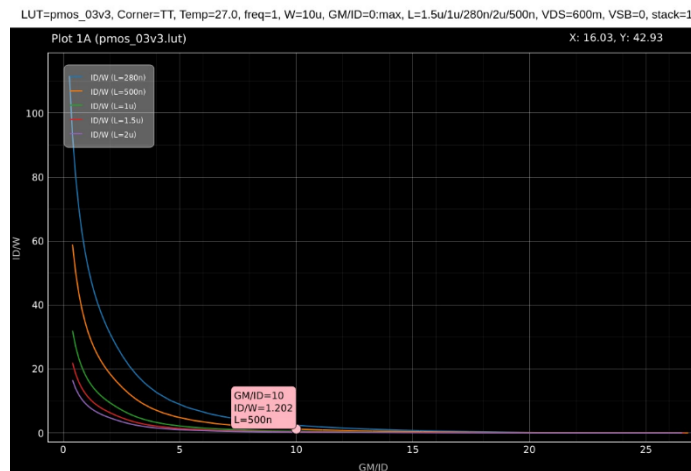


Figure 6: ID/W for tail current source



Using the first-order approximation $g_m/I_D = 2/V^*$, the transconductance efficiency for the tail current source was set to 10. A channel length of $L_5 = 0.5 \mu\text{m}$ (500 nm) was selected for this device. According to the ADT I_D/W chart for PMOS devices at this specific g_m/I_D and length, the current density is $I_D/W = 1.202 \text{ A/m}$. Given the total first-stage tail current requirement of $I_{tail} = 12.5 \mu\text{A}$, the width of the tail current source was calculated as follows:

$$W_5 = \frac{12.5 \mu\text{A}}{1.202 \text{ A/m}} = 10.4 \mu\text{m}$$

To generate this specific bias current, an external ideal $10 \mu\text{A}$ DC current source was utilized for the testbench bias circuitry. A diode-connected PMOS transistor (M_B) was designed to form a current mirror with M_5 . To accurately mirror and scale the $10 \mu\text{A}$ reference current to the required $12.5 \mu\text{A}$ tail current without altering the device's operating point, M_B was assigned the same channel length ($L_B = 0.5 \mu\text{m}$), while its width was scaled proportionally:

$$W_B = W_5 \times \left(\frac{10 \mu\text{A}}{12.5 \mu\text{A}} \right) = 10.4 \mu\text{m} \times 0.8 = 8.32 \mu\text{m}$$

1.3.4 Second-Stage Input Transistor Sizing (M_8)

The sizing of the second-stage NMOS input transistor (M_8) was determined by the Phase Margin (PM) requirement and the necessity to eliminate systematic offset. To achieve a stable Phase Margin ($\geq 70^\circ$), the non-dominant pole (ω_{p2}) must be pushed to a sufficiently high frequency. Following the suggested procedure, the non-dominant pole was placed at four times the unity gain frequency ($\omega_{p2} = 4\omega_u$).

Approximating the non-dominant pole as $\omega_{p2} \approx g_{m8}/C_L$, the required transconductance for M_8 was calculated:

$$g_{m8} = 4 \times \omega_u \times C_L = 4 \times 31.428 \text{ Mrad/s} \times 5 \text{ pF} = 628.5 \mu\text{S}$$

A critical design constraint in a two-stage OTA is avoiding systematic offset. This requires the DC voltage at the output of the first stage to perfectly bias M_8 such that $V_{GS8} = V_{GS4}$. To guarantee identical V_{GS} without relying on iterative simulation tweaks, M_8 was assigned the exact same channel length ($L_8 = 520 \text{ nm}$) and transconductance efficiency ($g_m/I_D = 15 \text{ V}^{-1}$) as the first-stage current mirror load (M_3, M_4).



Using the required transconductance and the fixed g_m/I_D ratio, the necessary drain current for the second stage (I_{D8}) was calculated:

$$I_{D8} = \frac{g_{m8}}{g_m/I_D} = \frac{628.5 \text{ uS}}{15 \text{ V}^{-1}} = 41.9 \text{ mA}$$

From the ADT charts for NMOS devices at $L = 520 \text{ nm}$ and $g_m/I_D = 15 \text{ V}^{-1}$, the current density is $I_D/W = 1.287 \text{ A/m}$. Finally, the width of the second-stage input transistor was calculated as:

$$W_8 = \frac{41.9 \text{ uA}}{1.287 \text{ A/m}} = 32.634 \text{ um}$$

1.3.5 Second-Stage Current Source Load (M_7)

The PMOS current source load for the second stage (M_7) is responsible for supplying the required 42 μA bias current to the second-stage input transistor (M_8). According to the suggested design procedure, this device forms a current mirror with the main bias transistor (M_B) and the first-stage tail current source (M_5), and consequently must share the same channel length. Therefore, the channel length was set to $L_7 = 0.5 \text{ mum}$.

With the channel length and gate-to-source voltage fixed by the primary biasing network, the width of M_7 was computed strictly through direct current scaling. Utilizing the reference bias current ($I_{bias} = 10 \text{ muA}$) and its corresponding device width ($W_B = 8.32 \text{ mum}$), the required width W_7 is calculated as follows:

$$W_7 = W_B \times \frac{I_{D7}}{I_{bias}} = 8.32 \text{ mum} \times \frac{42 \text{ muA}}{10 \text{ muA}} = 34.944 \text{ mum}$$

1.3.6 Nulling Resistor (VCR) Sizing (M_{10})

To ensure stability and improve the Phase Margin, a Voltage Controlled Resistor (VCR) implemented via an NMOS transistor (M_{10}) operating in the deep triode region was introduced in series with the Miller compensation capacitor (C_C). The primary objective of this device is to push the Right-Half-Plane (RHP) zero to infinity. According to the design procedure, this is achieved by setting the equivalent on-resistance of the triode transistor (R_z) equal to the inverse of the second-stage input transconductance ($1/g_{m8}$).

Equating the triode resistance expression of M_{10} with $1/g_{m8}$:

$$R_z = \frac{1}{g_{m8}}$$

$$\frac{1}{\mu_n C_{ox} \left(\frac{W}{L}\right)_{10} (V_{GS10} - V_{th})} = \frac{1}{\mu_n C_{ox} \left(\frac{W}{L}\right)_8 (V_{GS8} - V_{th})}$$

By canceling out the common process parameters ($\mu_n C_{ox}$), the sizing relationship simplifies to:

$$\left(\frac{W}{L}\right)_{10} (V_{GS10} - V_{th}) = \left(\frac{W}{L}\right)_8 (V_{GS8} - V_{th})$$

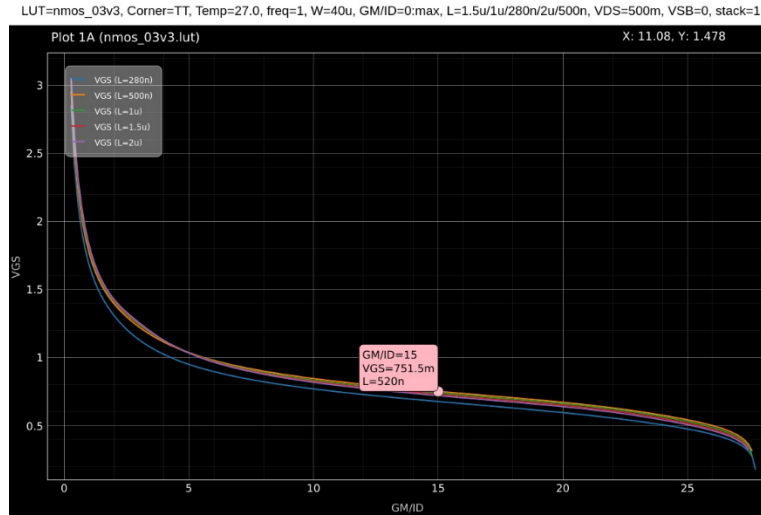


Figure 7: V_{GS8}

The gate of M_{10} is tied to V_{DD} (1.8 V) to ensure it remains strongly in the linear region, while V_{GS8} is defined by the DC bias point established by the first stage (751.5 mV). Substituting the known aspect ratio of the second-stage input transistor ($\left[\frac{W}{L}\right]_8 = 62.757$) and the corresponding effective overdrive voltages extracted from the design charts, the required aspect ratio for M_{10} was determined.

To minimize parasitic capacitance contributions while maintaining matching characteristics, the channel length was set identically to the other critical NMOS devices ($L_{10} = 520$ nm). Solving the equation yields the final width for the VCR transistor:

$$W_{10} = 1.277 \text{ } \mu\text{m}$$



1.3.7 Final Transistor Sizing and DC Operating Parameters

The following table summarizes the dimensions and the calculated DC operating point parameters for all transistors in the Two-Stage Miller OTA, derived using the g_m/I_D methodology design curves

Transistor	Type	W (um)	L (nm)	gm/ID (V-1)	gm (uS)
M_1, M_2 (Input Pair)	PMOS	13.76	430	15	93.75
M_3, M_4 (Stage 1 Load)	NMOS	4.84	520	15	93.75
M_5 (Tail Current)	PMOS	10.40	500	10	125.00
M_B (Bias Reference)	PMOS	8.32	500	10	100.00
M_8 (Stage 2 Input)	NMOS	32.63	520	15	630.00
M_7 (Stage 2 Load)	PMOS	34.95	500	10	420.00
M_{10} (Nulling VCR)	NMOS	1.28	520	—	—

2 PART 2: Open-Loop OTA Simulation

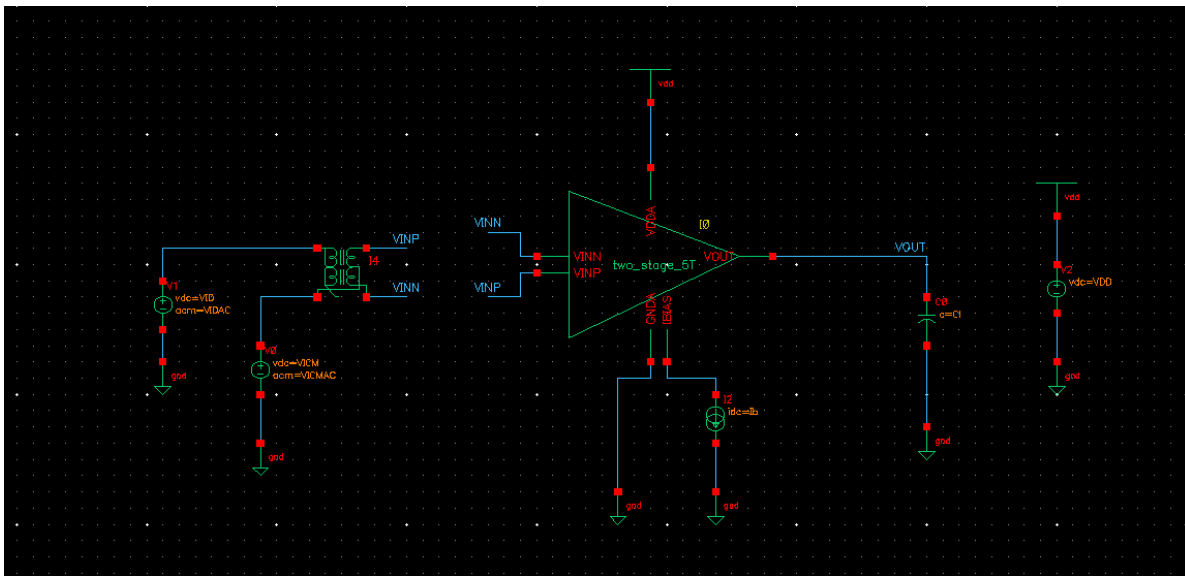


Figure 8: open loop OTA schematic



2.1 Schematic of the OTA and bias circuit with DC node voltages clearly annotated.

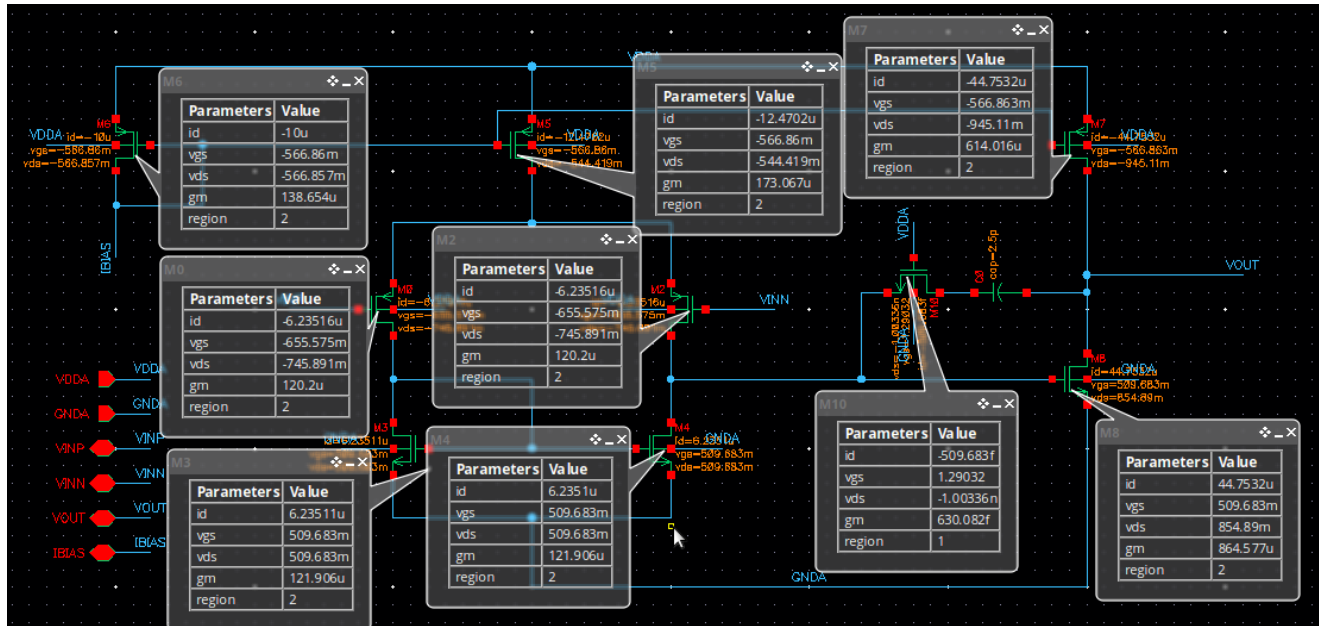


Figure 9:DC_OP

2.1.1 Is the current (and g_m) in the input pair exactly equal?

Yes, they are exactly equal. The simulation shows that both the left branch transistor (M_0) and the right branch transistor (M_2) carry an identical drain current of 6.235 μA and have an identical transconductance of 120.2 μS . This perfect match indicates that the systematic offset has been successfully eliminated by careful sizing. The drain voltages of both input transistors are identical ($V_{DS3} = V_{DS4} = 509.68 \text{ mV}$), meaning there is no channel-length modulation mismatch between the two branches.

2.1.2 What is DC voltage at the output of the first stage? Why?

The DC voltage at the output of the first stage (drain of M_4) is 509.68 mV. This voltage serves two purposes: first, it perfectly matches the drain voltage of the diode-connected load (M_3), which ensures the exact current mirroring and eliminates systematic offset. Second, it is precisely the gate-to-source voltage (V_{GS}) required by the second-stage input transistor (M_8) to sustain its designated bias current of 44.75 μA while operating in the saturation region.



2.1.3 What is DC voltage at the output of the second stage? Why?

The DC voltage at the final output is 854.89 mV. In an open-loop configuration with perfectly mitigated systematic offset, this output node floats to an intermediate equilibrium voltage dictated by the intersection of the $I - V$ output characteristics of the NMOS input transistor (M_8) and the PMOS load transistor (M_7). Since both devices are properly sized to carry the exact same current (44.75 μA) and both have identical channel lengths, the node naturally settles near $V_{DD}/2$, keeping both M_7 and M_8 strictly in the saturation region (Region 2).

2.2 Diff small signal ccs:

2.2.1 parameters (A_o , A_o in dB, BW, GBW, UGF).

Test	Output	Nominal
mini_project_1:two_stage_5T_tb1:1	A_o	2.816k
mini_project_1:two_stage_5T_tb1:1	A_o_{db}	68.99
mini_project_1:two_stage_5T_tb1:1	BW	2.603k
mini_project_1:two_stage_5T_tb1:1	fu	7.191M
mini_project_1:two_stage_5T_tb1:1	GBW	7.329M

Figure 10: A_o , A_o in dB, BW, GBW, UGF

2.2.2 Plot diff gain (in dB) vs frequency.

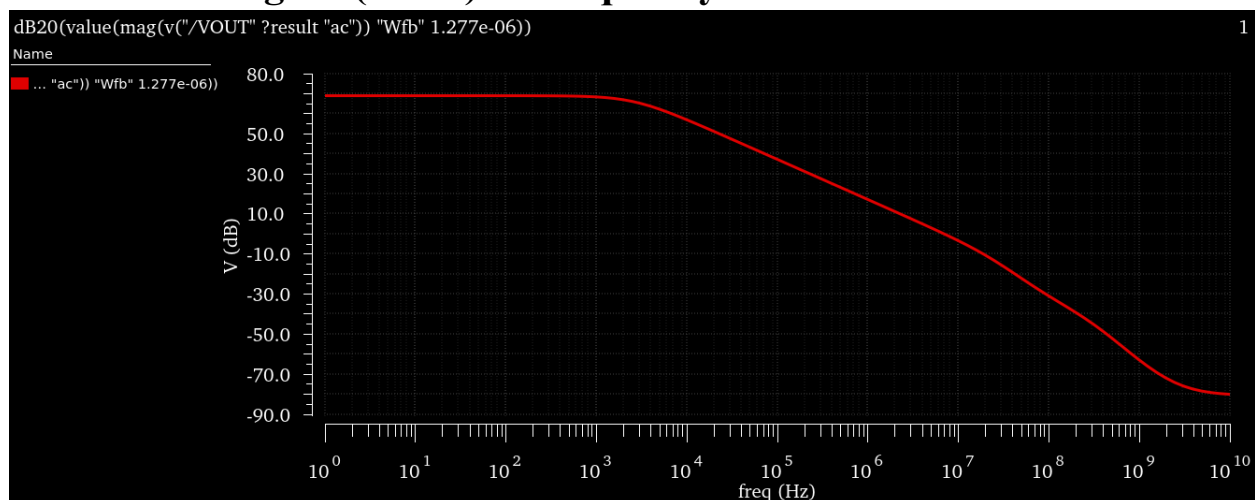


Figure 11: diff gain (in dB) vs frequency.



2.2.3 Comparison Between Simulation and Hand Calculations

The following table compares the small-signal parameters obtained from the AC simulation with the theoretical values derived during the hand-calculation phase.

Parameter	Hand Calculation (Theory)	Simulation Result (Cadence)	Comment / Justification
DC Gain (A_o)	> 2000 V/V (Target)	2816 V/V	Meets the $\leq 0.05\%$ static error target.
DC Gain in dB ($A_{o,dB}$)	> 66 dB (Target)	68.99 dB	Sufficient intrinsic gain achieved.
GBW	5.968 MHz	7.329 MHz	Simulated GBW is higher due to a higher actual g_m in the input pair.
Unity Gain Freq (f_u)	≈ 5.00 MHz	7.191 MHz	Matches the trend of the increased GBW.
Bandwidth (BW)	≈ 2.98 kHz	2.603 kHz	Consistent with the relationship $BW = GBW / A_o$.



2.3 CM small signal ccs:

2.3.1 Plot CM gain in dB vs frequency.

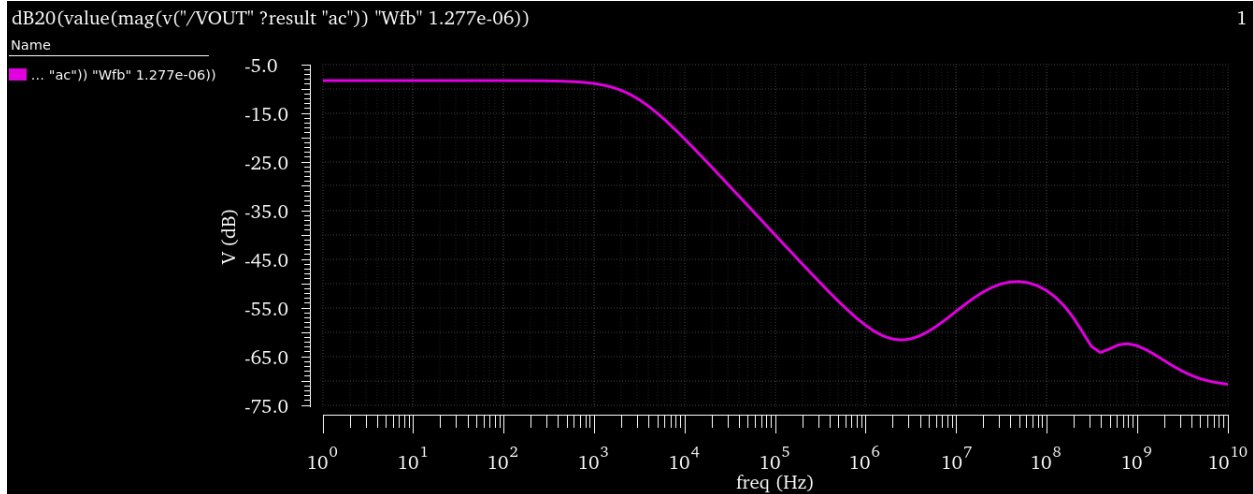


Figure 12: CM gain in dB vs frequency

The AC simulation for the common-mode gain (A_{cm}) was performed by applying an AC magnitude of 1 V to both inputs simultaneously ($VICMAC = 1, VIDAC = 0$). Based on the provided Bode plot, the DC common-mode gain settles at approximately -8.2 dB in the low-frequency flat band.

To compare this with your hand calculations, we derive the theoretical target for A_{cm} using the specifications provided in the project. The Common-Mode Rejection Ratio (CMRR) must be ≥ 74 dB, and the minimum target differential gain (A_d) derived from the static error was 66 dB. Since $CMRR_{dB} = A_{d,dB} - A_{cm,dB}$, the maximum allowable common-mode gain is theoretically calculated as:
$$A_{cm(max)} = A_{d(min)} - CMRR_{(min)} = 66 \text{ dB} - 74 \text{ dB} = -8 \text{ dB}$$



2.3.2 Comparison Between Simulation and Hand Calculations (CM Gain)

Parameter	Hand Calculation (Target)	Simulation Result (Cadence)	Comment / Justification
DC CM Gain (A_{cm})	≤ -8 dB	≈ -8.2 dB	Common-mode signals are successfully attenuated (gain < 0 dB).
CMRR	≥ 74 dB (Spec)	77.19 dB	The simulated CMRR safely exceeds the minimum project specification.

2.4 CMRR:

2.4.1 Plot CMRR in dB vs frequency.

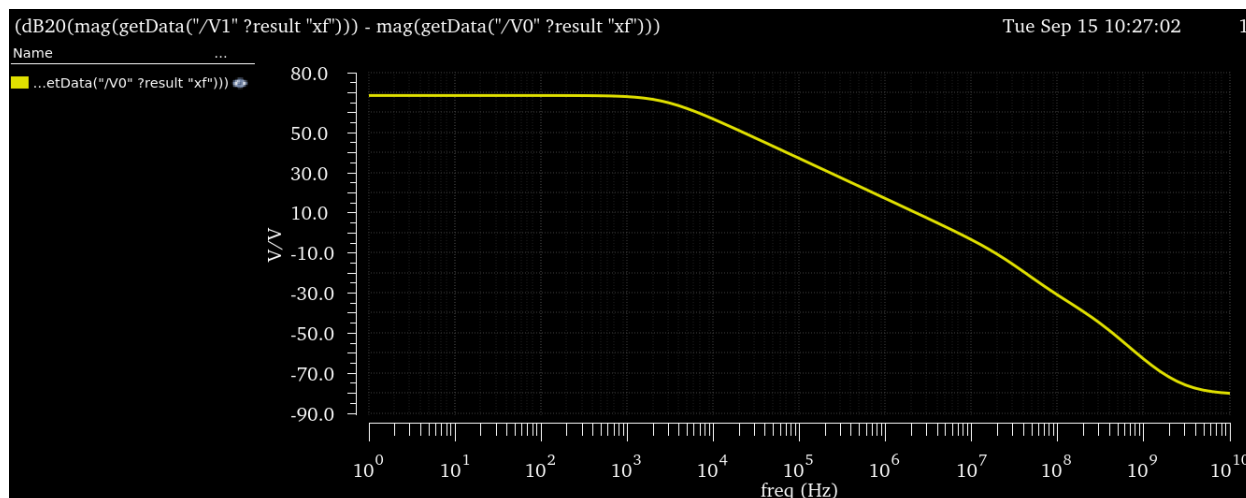


Figure 13: CMRR

To accurately evaluate the Common-Mode Rejection Ratio (CMRR) across the frequency spectrum, an XF (Transfer Function) analysis was performed. Unlike a standard AC analysis, the XF analysis efficiently computes the transfer functions from multiple independent sources to a single designated output node simultaneously. The CMRR curve was generated by evaluating the difference



between the differential transfer function and the common-mode transfer function in decibels. As observed in the generated Bode plot, the CMRR remains flat at low frequencies, settling at approximately 77.2 dB. This successfully meets and exceeds the project's minimum specification of 74 dB. At higher frequencies, the CMRR curve exhibits a noticeable roll-off. This degradation is an expected behavior in differential amplifiers, primarily caused by the parasitic capacitances at the common tail node of the first stage. These parasitics reduce the effective impedance of the tail current source at high frequencies, thereby increasing the common-mode gain and deteriorating the overall CMRR.

2.4.2 Comparison Between Simulation and Hand Calculations (CMRR)

Parameter	Target Specification	Simulation Result (XF Analysis)	Comment / Justification
CMRR @ DC	≥ 74 dB	≈ 77.2 dB	Safely exceeds the minimum requirement, indicating an excellent design of the tail current source (M_5) output resistance.



2.5 Diff large signal ccs:

2.5.1 Plot V_{OUT} vs V_{ID} .

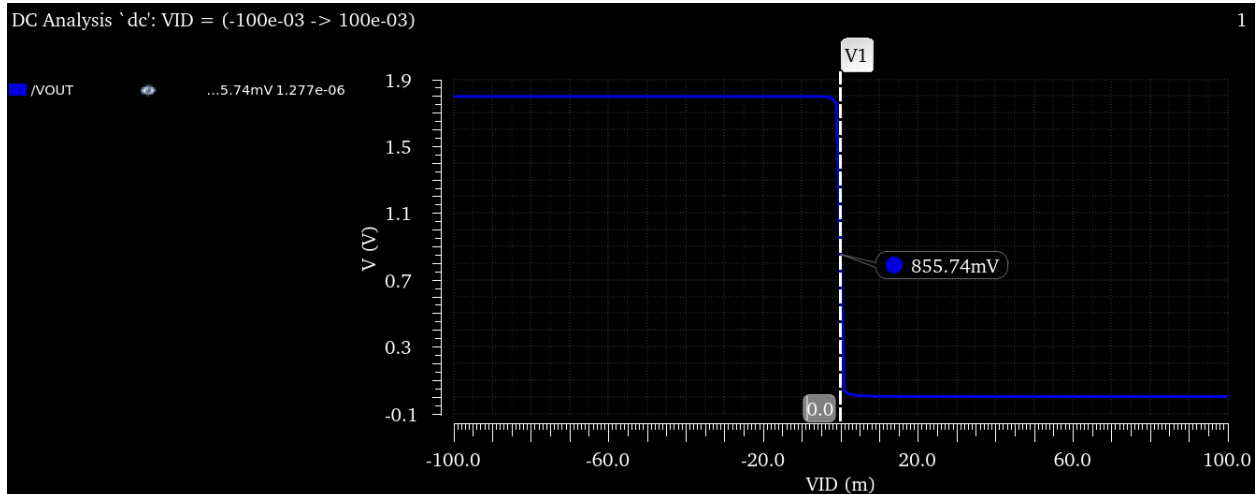


Figure 14: V_{OUT} vs. V_{ID}

2.5.2 Plot the derivative of V_{OUT} vs V_{ID} .

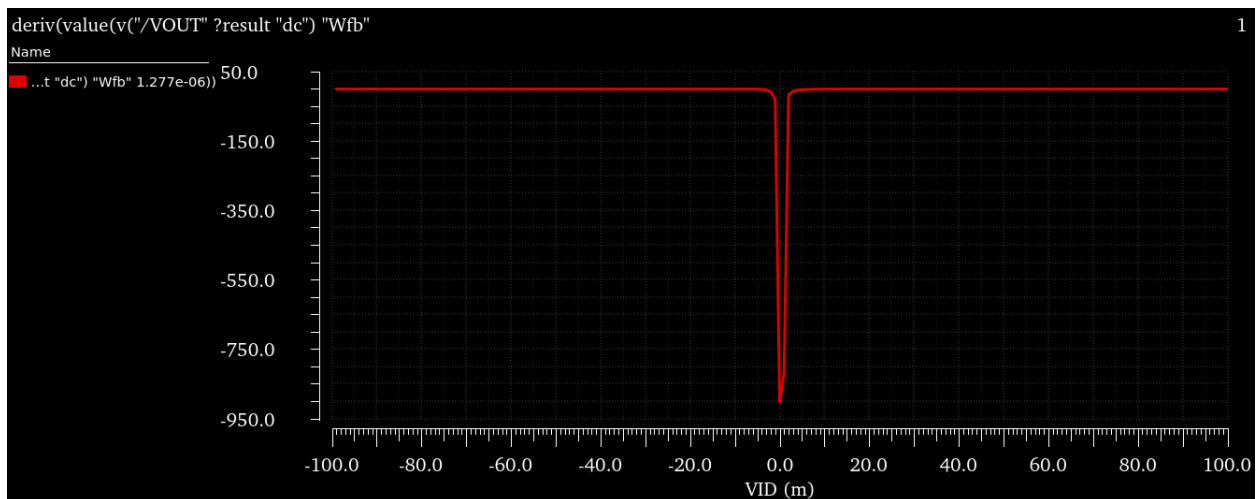


Figure 15: derivative of V_{OUT} vs V_{ID}

2.5.3 V_{out} at $V_{ID} = 0$.

$V_{out} = 855.74$ mV



2.6 CM large signal ccs (region vs VICM):

2.6.1 Plot “region” OP parameter vs VICM for the input pair and the tail current source.

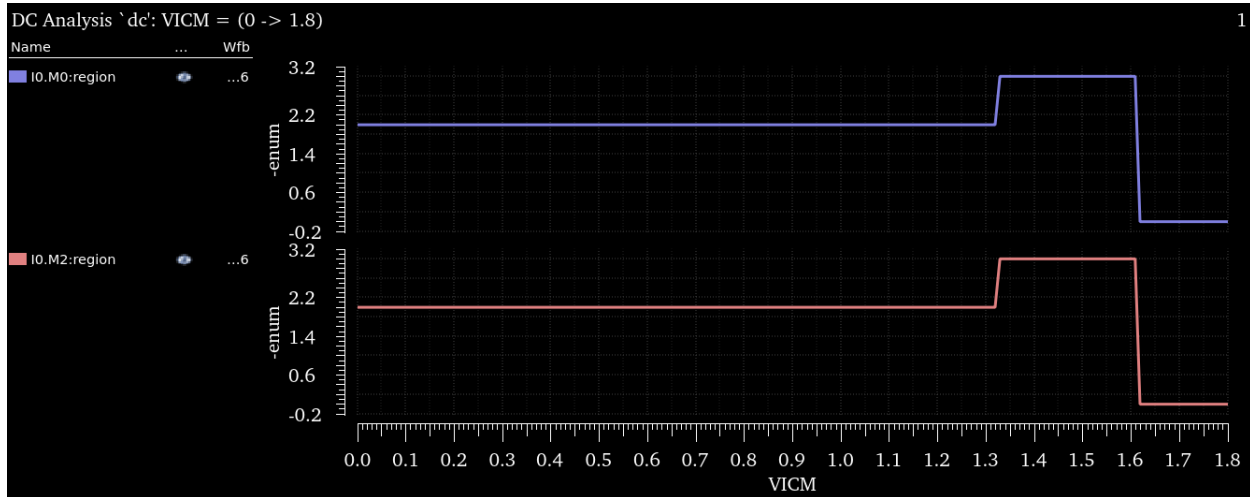


Figure 16: input pair CMIR

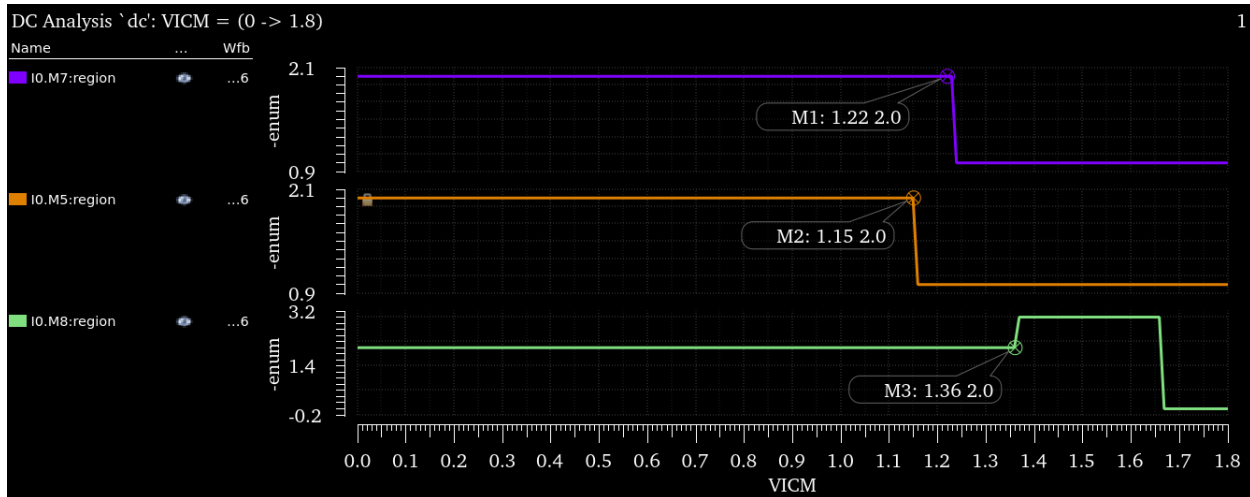


Figure 17: the tail current source Regions

2.6.2 input range (CMIR).

CMIR_low = 0 v

CMIR_high = 1.15v



2.7 CM large signal ccs (GBW vs VICM):

2.7.1 Plot GBW vs VICM.

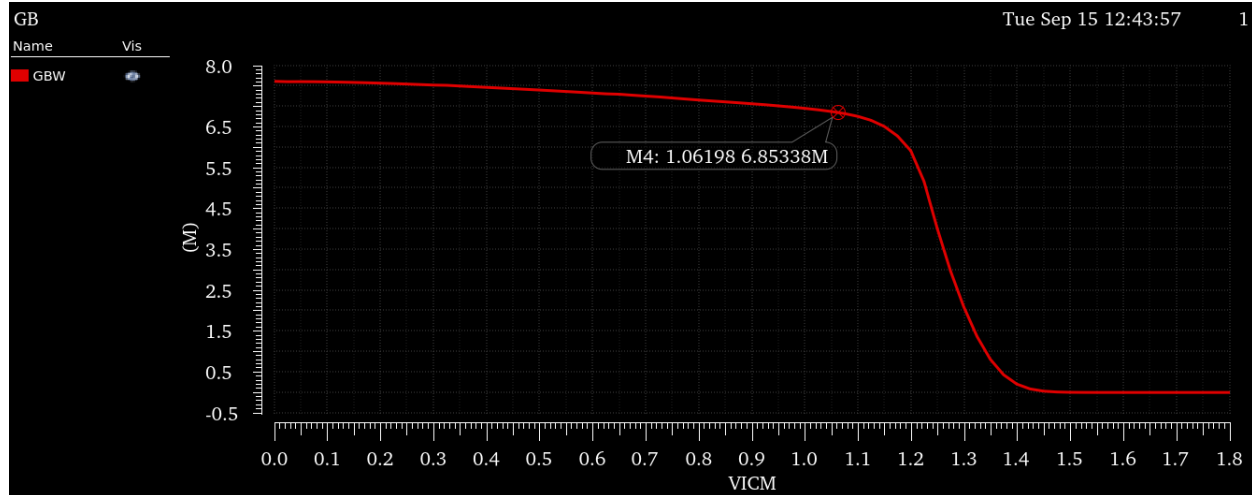


Figure 18:GBW vs. VICM

2.7.2 Calculate the input range

→ the effective Common-Mode Input Range (CMIR) spans from 0 V to 1.06 V

→ This successfully meets the project requirements ($CMIR_{low} \leq 0.2$ V and $CMIR_{high} \geq 1.0$ V).



3 PART 3: Closed-Loop OTA Simulation

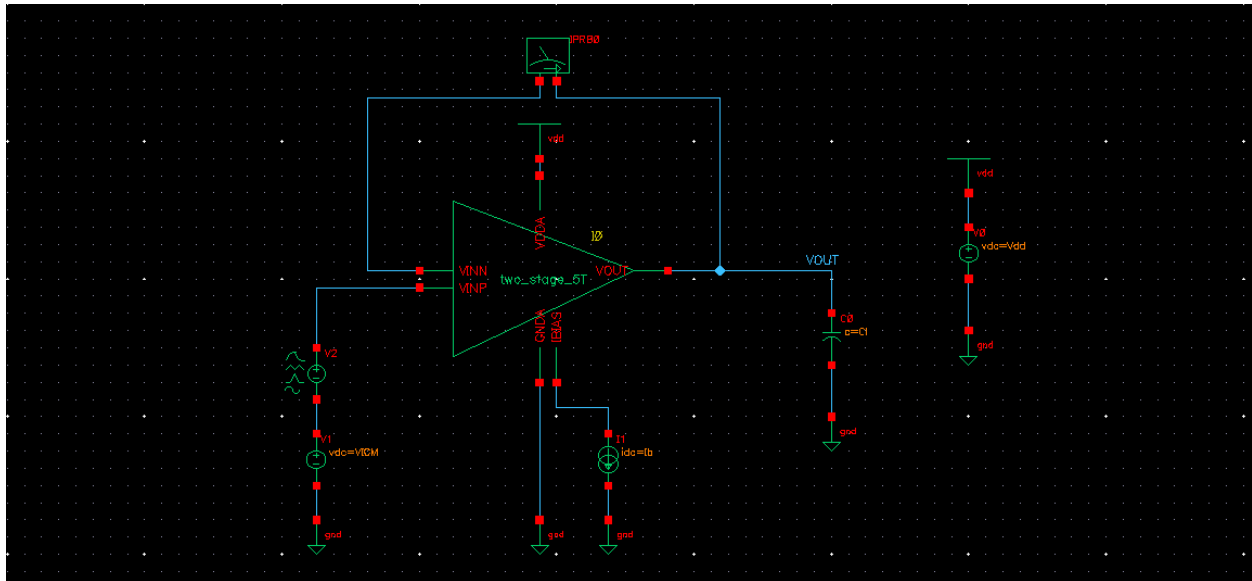


Figure 19: Closed-Loop OTA Simulation

3.1 Schematic of the OTA and the bias circuit with DC OP point

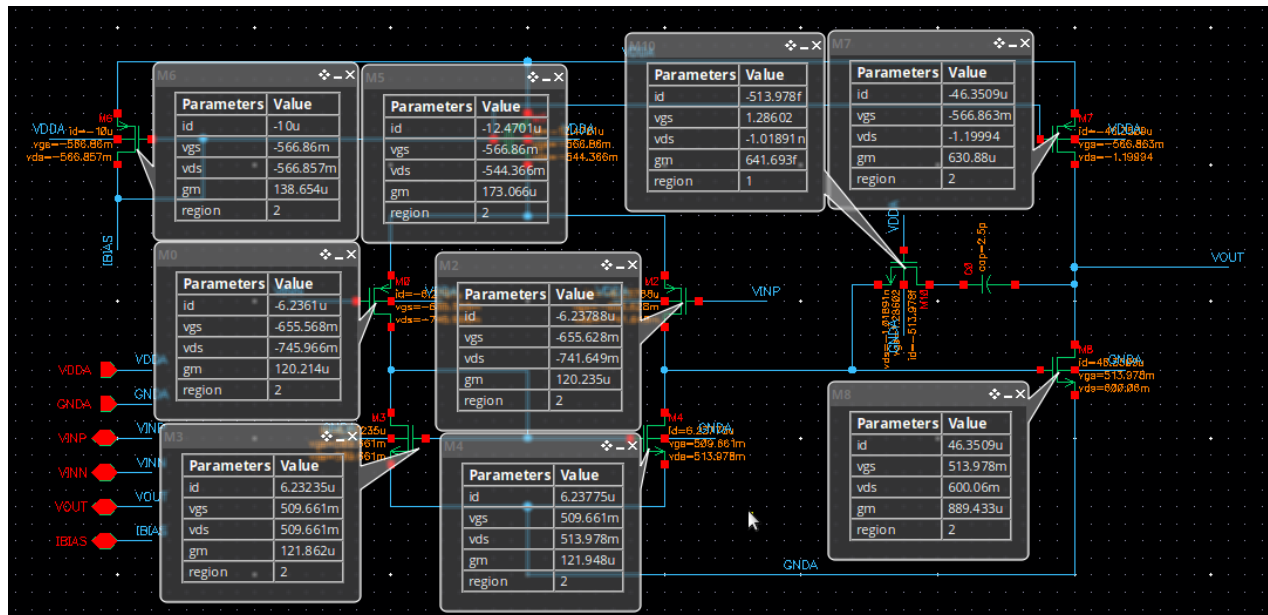


Figure 20: DC OP point Closed loop



3.1.1 Are the DC voltages at the input terminals of the op-amp exactly equal? Why?

Yes, they are approximately equal (~600 mV). This is due to the negative feedback in the unity-gain buffer configuration, which creates a "virtual short" forcing the inverting terminal to closely track the non-inverting terminal.

3.1.2 Is the DC voltage at the output of the first stage exactly equal to the value in the open-loop simulation? Why?

In the closed-loop state, the voltage settles precisely at 513.9 mV. The negative feedback automatically compensates for any systematic offset, adjusting this node to provide the exact V_{GS} required by M_8 to keep the devices in saturation.

3.1.3 Is the current (and gm) in the input pair exactly equal? Why?

They are extremely close but not 100% identical ($I_{D0} = 6.236 \mu\text{A}$ and $I_{D2} = 6.237 \mu\text{A}$). This minor mismatch is caused by the different drain voltages across the two transistors, which introduces a slight channel-length modulation effect.

3.2 Loop gain:

3.2.1 Plot loop gain in dB and phase vs frequency.

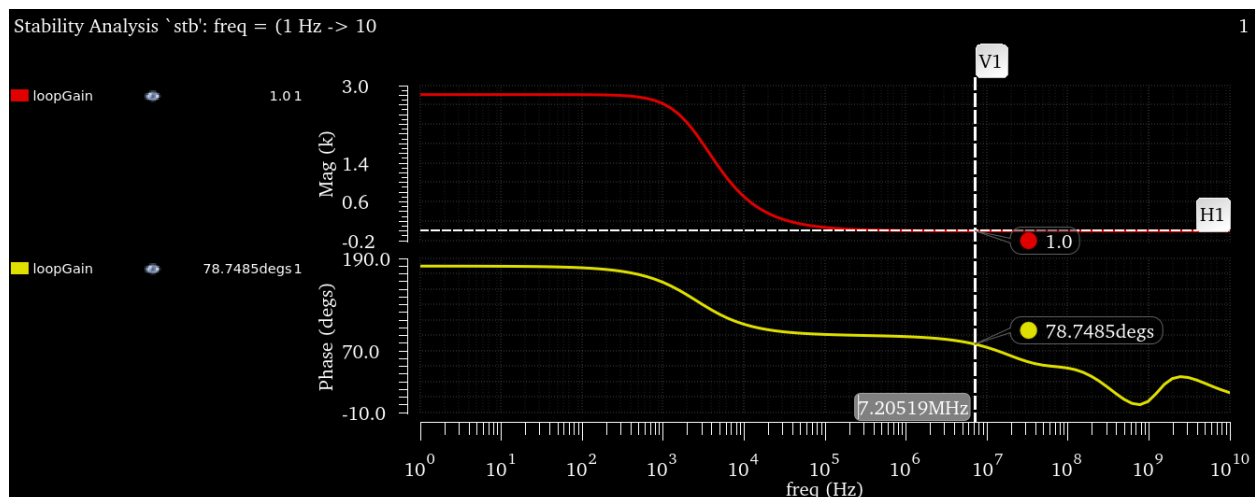


Figure 21: Loop gain and phase



3.2.2 Compare DC gain, f_u , and GBW with those obtained from open-loop simulation. Comment

Test	Output	Nominal
mini_project_1:two_stage_5T_tb2:1	Ao	2.814k
mini_project_1:two_stage_5T_tb2:1	f_u	7.205M
mini_project_1:two_stage_5T_tb2:1	GBW	7.335M

Figure 22:DC gain, f_u , and GBW

3.2.3 Loop Gain Comparison (STB vs. Open-Loop)

Parameter	Open-Loop (AC Analysis)	Closed-Loop (STB Analysis)
DC Gain (A_o)	2.816 k	2.814 k
Unity Gain Freq (f_u)	7.191 MHz	7.205 MHz
GBW	7.329 MHz	7.335 MHz

Comment:

The parameters obtained from the STB analysis in the closed-loop configuration are virtually identical to the open-loop AC analysis. This demonstrates that the unity-gain buffer configuration effectively stabilizes the DC operating point without altering the core small-signal characteristics of the OTA.

3.2.4 Report PM. Compare with hand calculations. Comment.

- **Reported PM:** The simulated Phase Margin from the STB analysis is 78.75° .
- **Comparison with Hand Calculations:** During the hand calculation phase, the non-dominant pole was designed to be at four times the unity-gain frequency ($\omega_{p2} = 4\omega_u$). Assuming the Right-Half-Plane (RHP) zero is perfectly cancelled by the nulling resistor, the theoretical expected Phase Margin is approximately 76° (calculated as $180^\circ - 90^\circ - \arctan(1/4)$).



- **Comment:** The simulated result (78.75°) aligns closely with the theoretical calculation and safely exceeds the minimum target specification of $\geq 70^\circ$. This confirms that the VCR nulling resistor (M_{10}) was sized correctly and effectively eliminated the RHP zero, resulting in a highly stable closed-loop system.

3.2.5 Comparison Between Simulation and Hand Calculations (Loop Gain)

Parameter	Hand Calculation (Theory/Target)	Simulation Result (STB)	Comment
DC Loop Gain (A_o)	> 66 dB (Spec)	68.99 dB	Safely meets the required intrinsic gain target for the desired static error.
Phase Margin (PM)	$\approx 76^\circ$ (Theoretical)	78.75°	Closely matches the expected value, confirming successful RHP zero cancellation.
Unity Gain Freq (f_u)	≈ 5.00 MHz (Target)	7.205 MHz	The simulated bandwidth is higher due to a slightly larger effective g_m in the input pair.



3.3 Slew rate:

3.3.1 Report Vin and Vout overlaid.

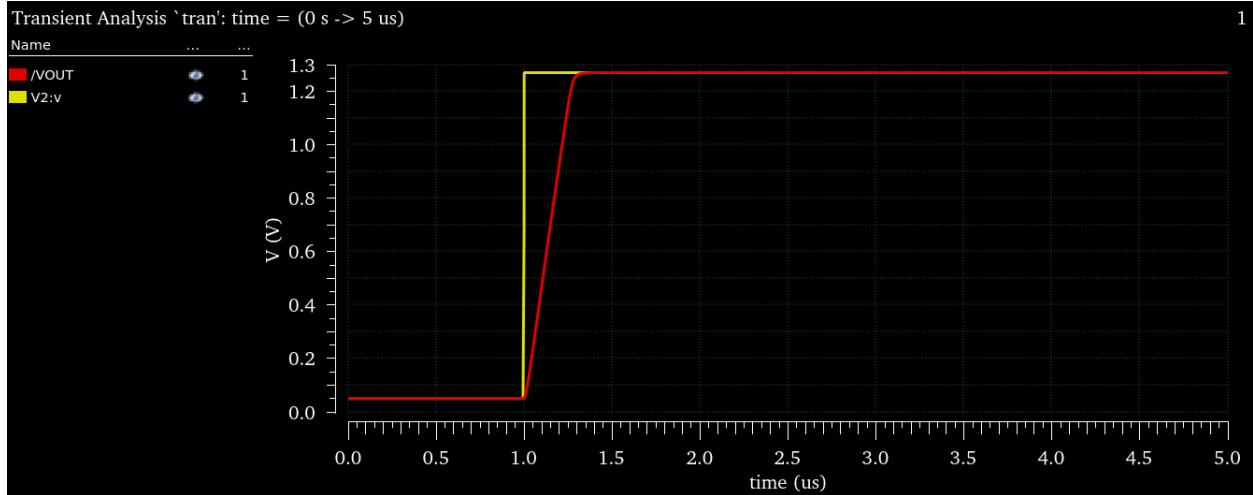


Figure 23: Vin and Vout overlaid

3.3.2 Report the slew rate.

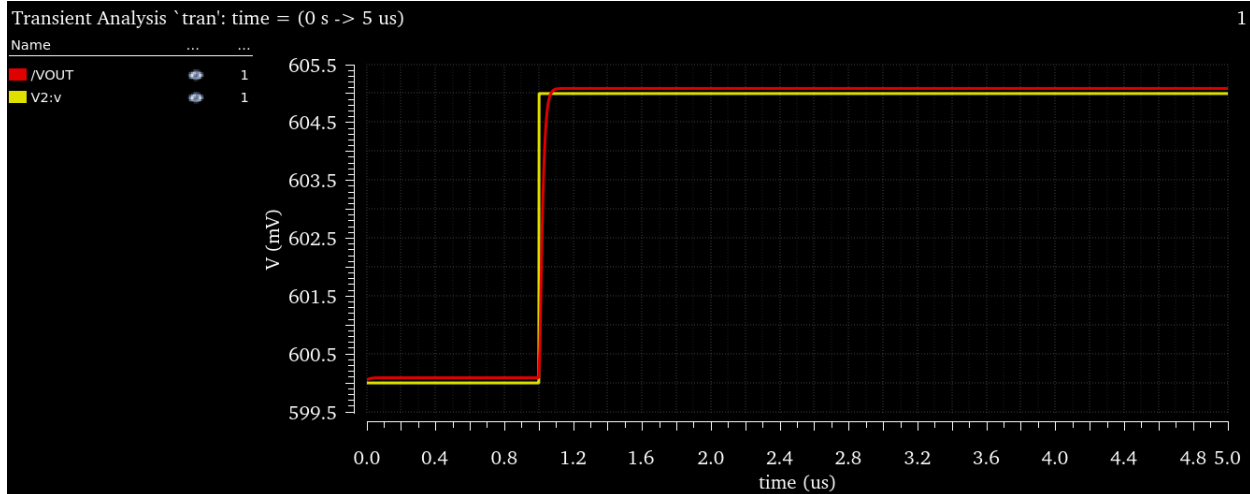
Path	Signal	Value
mini_project_1:two_stage_5T_tb2:1	SR	4.542M

Comment:

The simulated SR is slightly lower than the ideal theoretical value. This is expected in practical designs due to parasitic capacitances at internal nodes (e.g., the output of the first stage and the tail node) which absorb a fraction of the available charging current during large-signal transitions.

3.4 Settling time:

3.4.1 Calculate the output rise time from simulation.



Test	Output	Nominal
mini_project_1:two_stage_5T_tb2:1	trise	37.02n
mini_project_1:two_stage_5T_tb2:1	settling time	1.05u

Figure 24: the output rise time

3.4.2 Do you see any ringing? Why?

- **Observation:** No, there is absolutely no ringing or overshoot observed in the output transient response. The output voltage rises smoothly and settles directly at its final value.
- **Justification:** This highly stable, well-damped time-domain response is a direct consequence of the excellent Phase Margin (78.75°) achieved in the design. Closed-loop systems with a Phase Margin significantly greater than 60° are sufficiently damped and do not exhibit ringing. This practically proves that the nulling resistor (M_{10}) was properly sized and successfully eliminated the RHP zero, ensuring robust stability.